MonoArt: Progressive Structural Reasoning for Monocular Articulated 3D Reconstruction

Haitian Li*, Haozhe Xie*, Junxiang Xu, Beichen Wen, Fangzhou Hong, and Ziwei Liu ✉

S-Lab, Nanyang Technological University, 637335 Singapore
https://lihaitian.com/MonoArt

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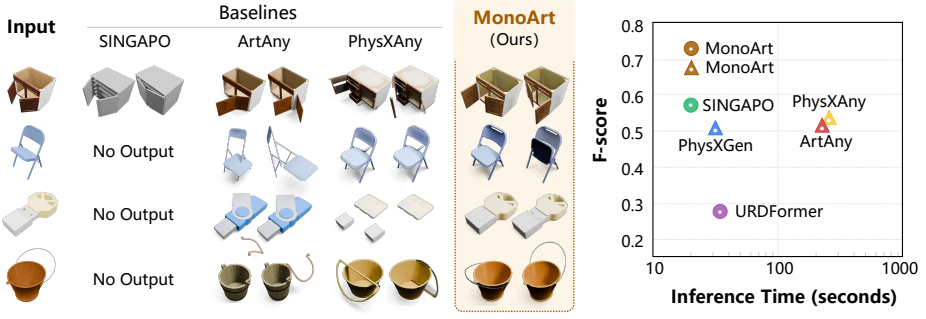

Fig. 1: (Left) Qualitative results of SINGAPO [24], Articulate-Anything (ArtAny) [18], PhysX-Anything (PhysXAny) [4], and MonoArt on diverse objects. **(Right)** F-score vs. inference time on the PartNet-Mobility [51] test set. Circles indicate models evaluated on 7 categories, while triangles denote models supporting all 46 categories.

Abstract. Reconstructing articulated 3D objects from a single image requires jointly inferring object geometry, part structure, and motion parameters from limited visual evidence. A key difficulty lies in the entanglement between motion cues and object structure, which makes direct articulation regression unstable. Existing methods address this challenge through multi-view supervision, retrieval-based assembly, or auxiliary video generation, often sacrificing scalability or efficiency. We present MonoArt, a unified framework grounded in progressive structural reasoning. Rather than predicting articulation directly from image features, MonoArt progressively transforms visual observations into canonical geometry, structured part representations, and motion-aware embeddings within a single architecture. This structured reasoning process enables stable and interpretable articulation inference without external motion templates or multi-stage pipelines. Extensive experiments on PartNet-Mobility demonstrate that MonoArt achieves state-of-the-art performance in both reconstruction accuracy and inference speed. The framework further generalizes to robotic manipulation and articulated scene reconstruction.

Keywords: Monocular Articulated 3D Reconstruction · Progressive Structural Reasoning · Kinematic Estimation

* Equal Contribution ✉ Corresponding Author

1 Introduction

Reconstructing articulated 3D objects from visual observations remains a fundamental challenge in computer vision and graphics. Such objects (*e.g.*, laptops, cabinets) are ubiquitous in daily environments, and generating high-fidelity, interactable 3D assets at scale is increasingly vital for applications in robotics [1, 2, 55] and scene synthesis [49, 53, 54, 58]. Despite recent progress in 3D object generation [10, 17, 46, 52], producing high-quality articulated assets still requires extensive manual effort. Unlike rigid reconstruction, articulated reconstruction demands understanding both an object's structural composition and the kinematic relationships among its parts.

Articulated 3D object modeling has attracted increasing attention in recent years, driving significant advances in reconstruction. Existing approaches can be broadly classified into three categories. Most existing methods [13, 25, 34, 48, 50] rely on multiple images captured from videos or frame sequences indicating different articulation states (*e.g.*, open and closed). While achieving higher reconstruction accuracy, they require multiple motion states of the same object, which are not always readily available in practice. To relax this constraint, subsequent approaches [8, 18, 24] take a single image as input but reconstruct 3D objects by retrieving and assembling parts from pre-built asset libraries, often leading to texture misalignment and geometric inaccuracies. Very recent works move beyond retrieval and attempt single-image reconstruction of articulated objects by generating auxiliary videos [31], leveraging vision–language models [4], or relying on predefined motion directions to infer articulation cues [12, 21]. While these approaches demonstrate the potential of single-image articulation modeling, the former is complex and computationally expensive, whereas the latter depends on handcrafted priors that limit generalization. This reliance on external cues reflects the lack of intrinsic 3D understanding, making it difficult to infer part composition and spatial relationships directly from a single image.

To address these limitations, we propose **MonoArt**, an end-to-end framework for monocular articulated 3D reconstruction grounded in progressive structural reasoning. Instead of directly regressing articulation parameters from image features, MonoArt progressively constructs structured part representations and lifts them into motion-aware embeddings, enabling stable and interpretable kinematic prediction. Specifically, **TRELLIS-based 3D Generator** first produces a canonical 3D shape from the input image, providing a stable geometric foundation. **Part-aware Semantic Reasoner** then lifts geometry-aligned point features into globally contextualized part-level embeddings through triplane aggregation and transformer-based refinement under 3D structural supervision. These part-aware features are subsequently processed by a **Dual-Query Motion Decoder**, which decouples spatial motion anchors and semantic part representations, and performs iterative refinement to reason about component-level motion patterns. Finally, **Kinematic Estimator** predicts part-level articulation parameters, including the part centroid, mask, joint type, axis, origin, and motion limits, and infers the kinematic tree structure, yielding a coherent articulated 3D reconstruction. As illustrated in Figure 1, MonoArt achieves both

higher F-score and substantially lower inference time than existing approaches, demonstrating that explicit structural priors enable not only more accurate but also more efficient articulated reconstruction.

The contributions are summarized as follows:

- We show that embedding 3D structural priors simplifies single-image articulated reconstruction by eliminating reliance on video generation, handcrafted motion templates, or vision–language priors, enabling reliable part decomposition and articulation reasoning.
- We propose MonoArt, a unified end-to-end framework that progressively reasons from geometry to kinematics, disentangling shape recovery, part-aware encoding, motion decoding, and kinematic regression for stable and physically meaningful articulation inference.
- MonoArt achieves state-of-the-art performance on PartNet-Mobility in both geometric and articulation metrics while significantly reducing inference time. The framework further generalizes to robotic manipulation and indoor 3D reconstruction tasks.

2 Related Work

Articulated Object Modeling. Articulated object modeling aims to recover object geometry together with part structure and motion from visual observations. Early multi-view methods [44, 48] adopt neural implicit representations to model category-level articulated shapes as deformations of a canonical template, enabling view synthesis but without explicit part or kinematic modeling. Later works introduce explicit part decomposition and motion reasoning. PARIS [25] aligns reconstructions across articulation states to estimate rigid parts and transformations. DTA [50] and ArticulatedGS [13] jointly model geometry, segmentation, and articulation parameters from multi-view RGB-D or Gaussian Splatting, producing motion-ready digital twins with high geometric fidelity. More recent approaches reduce input requirements by leveraging generative or language priors. FreeArt3D [7] optimizes geometry from sparse articulated views, while SINGAPO [24], NAP [19], and MeshArt [11] predict articulation trees and synthesize articulated parts. Articulate-Anything [18] formulates the task as vision–language reasoning to infer symbolic part hierarchies, and PhysX-Anything [4] further incorporates VLM priors to predict physically plausible structures and interactions. These methods improve scalability and controllability, often at the expense of precise instance-level reconstruction.

3D Part Segmentation. 3D part segmentation aims to decompose objects into semantic parts. Early learning-based methods [14, 22, 23, 30, 38–40, 61] focus on fully supervised point- or mesh-level classification using curated 3D datasets [6, 9, 35, 60]. While effective, these approaches are limited by the scale and diversity of part annotations. To improve open-world generalization and enable zero-shot capabilities, recent methods leverage 2D foundation vision models [16, 20, 29, 37, 41]. PartSLIP [28], PartSLIP++ [62], and ZeroPS [57] transfer image–language and segmentation priors to 3D via multi-view reasoning or

prompt-based inference, while PartDistill [47], SaMesh [45], and SAMPart3D [59] distill 2D foundation features into geometric representations. More recently, scaling-based approaches train feed-forward 3D segmentation models with large-scale part annotations. Find3D [33] leverages foundation models for pseudo-label generation, PartField [26] learns ambiguity-aware continuous feature fields, and P3-SAM [32] and PartSAM [63] demonstrate that large-scale part supervision yields strong point-wise representations for part prompting.

3 Our Approach

3.1 Overview

MonoArt reconstructs articulated 3D objects from a single image by progressively transforming visual observations into geometry-aware, part-aware, and motion-aware representations. As shown in Fig. 2, the framework consists of four components: TRELLIS-based 3D Generator (Sec. 3.2), Part-Aware Semantic Reasoner (Sec. 3.3), Dual-Query Motion Decoder (Sec. 3.4), and Kinematic Estimator (Sec. 3.5).

Given an input image $\mathbf{I}$, TRELLIS-based 3D Generator reconstructs a canonical geometry $\mathbf{O}$ using a frozen TRELLIS backbone and produces geometry-aligned latent features $\mathbf{Z}$. Based on $(\mathbf{O}, \mathbf{Z})$, Part-Aware Semantic Reasoner derives part-aware features $\mathbf{H}$ that encode explicit part decomposition guided by 3D structural annotations. Then, Dual-Query Motion Decoder initializes position and content queries $(\mathbf{Q}_p^0, \mathbf{Q}_c^0)$ from $(\mathbf{H}, \mathbf{F}_{\text{geo}})$ and iteratively refines them to obtain motion-aware representations $(\mathbf{Q}_p^L, \mathbf{Q}_c^L)$, jointly reasoning about part localization and motion semantics. Finally, Kinematic Estimator transforms the refined queries into explicit articulation parameters, namely part masks $\mathbf{m}_m$, motion types $\mathbf{m}_t$, motion origins $\mathbf{m}_o$, motion axes $\mathbf{m}_a$, and motion limits $\mathbf{m}_l$, while inferring the kinematic hierarchy to produce an articulated 3D representation.

3.2 TRELLIS-based 3D Generator

Given a single RGB image $\mathbf{I}$, MonoArt first reconstructs a canonical 3D object geometry using TRELLIS [52] as a frozen 3D generation backbone. TRELLIS predicts a structured sparse voxel latent representation $\mathbf{Z} \in \mathbb{R}^{N_z \times N_z \times N_z \times d_1}$, where N_z denotes the voxel grid resolution and each active voxel stores a d_1-dimensional feature while empty regions remain inactive, forming a spatially structured sparse volume. The latent $\mathbf{Z}$ is subsequently decoded by the mesh decoder of TRELLIS to produce an explicit 3D mesh $\mathbf{O}$, which serves as the canonical geometry for downstream part reasoning and articulation inference.

3.3 Part-Aware Semantic Reasoner

Our goal is to derive part-aware point features $\mathbf{H}$ from the canonical geometry $\mathbf{O}$ and the sparse voxel latent $\mathbf{Z}$ predicted by TRELLIS-based 3D generator, so that downstream modules can reason about object parts and their motions.

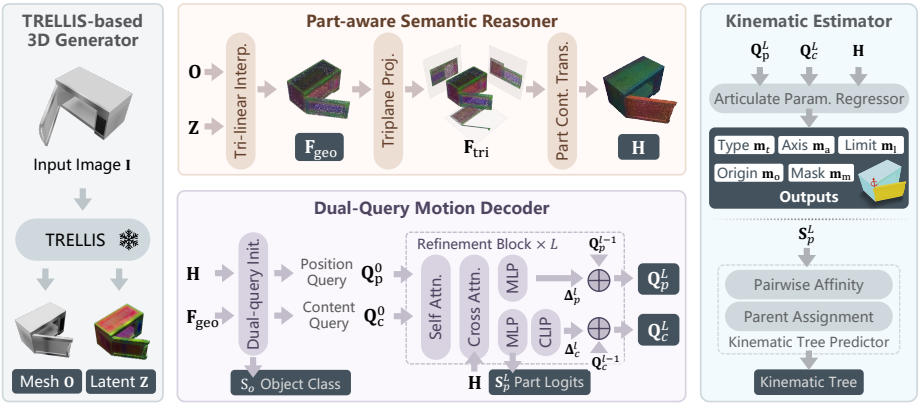

Fig. 2: Overview of MonoArt. TRELLIS-based 3D Generator reconstructs a canonical shape from a single image. Part-Aware Semantic Reasoner derives tri-plane-based part embeddings. Dual-Query Motion Decoder performs iterative motion reasoning, and Kinematic Estimator predicts part-level articulation parameters (motion type, origin, axis, limits) and infers the kinematic tree structure. Note that "Attn.", "Interp.", "Proj.", "Cont.", "Trans.", and "Init." represent "Attention", "Interpolation", "Projection", "Contrast", "Transformer", and "Initialization", respectively. $\oplus$ and $\otimes$ denote element-wise addition and matrix multiplication, respectively.

Tri-linear Interpolation. We sample M point sets $\{\mathbf{p}_m\}_{m=1}^{M}$ on the surface of $\mathbf{O}$, where $\mathbf{p}_m \in \mathbb{R}^3$. Given the sparse voxel feature volume $\mathbf{Z}$, we obtain point-aligned features by tri-linear interpolation over the neighboring voxels:

$$\mathbf{f}_m = \text{TrilinearInterp}(\mathbf{Z}, \mathbf{p}_m), \qquad \mathbf{F}_{\text{geo}} = \{\mathbf{f}_m\}_{m=1}^{M}. \tag{1}$$

Here, each point $\mathbf{p}_m$ is first mapped to its corresponding continuous voxel coordinate in the $N_z \times N_z \times N_z$ grid. The feature $\mathbf{f}_m \in \mathbb{R}^{d_1}$ is then computed as the weighted combination of the eight neighboring voxel features according to standard tri-linear interpolation [56]. This converts the discrete voxel latent $\mathbf{Z}$ into continuous surface-aligned features $\mathbf{F}_{\text{geo}} \in \mathbb{R}^{M \times d_1}$.

Triplane Projection. To incorporate global spatial context while preserving geometric structure, we project the geometry-aligned point features $\mathbf{F}_{\text{geo}}$ onto three orthogonal planes (XY, YZ, ZX) following the triplane formulation [5]. Specifically, each surface point $\mathbf{p}_m$ is orthographically projected onto the three planes according to its 3D coordinates, and its feature $\mathbf{f}_m$ is accumulated to the corresponding 2D grid locations via bilinear interpolation. This yields three feature maps of resolution $N_t \times N_t$, forming $\mathbf{F}_{\text{tri}} \in \mathbb{R}^{3 \times N_t \times N_t \times d_1}$.

Part Contrast Transformer. The triplane features $\mathbf{F}_{\text{tri}}$ are flattened into a token sequence $\mathbf{T} \in \mathbb{R}^{3N_t^2 \times d_1}$ and processed by a multi-layer self-attention Transformer to capture global interactions across planes, producing refined tokens $\mathbf{T}'$. The refined tokens are reshaped back into triplane feature maps $\mathbf{F}'_{\text{tri}} \in \mathbb{R}^{3 \times N_t \times N_t \times d_1}$. For each surface point $\mathbf{p}_m$, we project its 3D coordinate onto the three orthogonal planes and query the refined triplane feature maps via a

tri-plane sampling operation. This yields the updated point embedding

$$\mathbf{h}_m = \mathrm{MLP}(\mathrm{TriQuery}(\mathbf{F}'_{\mathrm{tri}}, \mathbf{p}_m)), \qquad \mathbf{H} = \{\mathbf{h}_m\}_{m=1}^{M}, \tag{2}$$

where TriQuery projects $\mathbf{p}_m$ onto the XY/YZ/ZX planes, bilinearly samples each plane feature map, and concatenates the three sampled features. Then, the MLP then lifts the aggregated feature from dimension d_1 to d_2. These embeddings encode part-aware structural representations and are supervised by 3D part annotations using triplet loss [43].

3.4 Dual-Query Motion Decoder

Articulated reasoning requires jointly modeling what constitutes a movable part and where its motion is spatially anchored. To this end, we adopt a dual-query formulation that disentangles semantic representation and geometric localization via two complementary query types: a content query $\mathbf{Q}_c \in \mathbb{R}^{N_q \times d_2}$ encoding part semantics, and a position query $\mathbf{Q}_p \in \mathbb{R}^{N_q \times 3}$ representing spatial motion anchors, where N_q denotes the number of dual queries. The dual queries are initialized from global object context and subsequently refined through L stacked refinement blocks in an iterative manner.

Dual-query Initialization. Given the part-aware point embeddings $\mathbf{H}$ and geometry-aligned features $\mathbf{F}_{\mathrm{geo}}$, we first aggregate global object context by applying global pooling over $\mathbf{H}$ and $\mathbf{F}_{\mathrm{geo}}$, followed by feature concatenation to obtain an object-level representation. This aggregated feature is projected to generate the initial content queries $\mathbf{Q}_c^0 \in \mathbb{R}^{N_q \times d_2}$ and position queries $\mathbf{Q}_p^0 \in \mathbb{R}^{N_q \times 3}$, together with an auxiliary object-category prediction S_o.

Refinement Block. The refinement block iteratively updates the dual queries over L layers to progressively refine articulation hypotheses. Given the initialized position query $\mathbf{Q}_p^0 \in \mathbb{R}^{N_q \times 3}$ and content query $\mathbf{Q}_c^0$, each layer first applies self-attention to model inter-part interactions, followed by cross-attention where the queries attend to the visual feature $\mathbf{H}$ to retrieve image evidence. For the l-th layer, the position query $\mathbf{Q}_p^l$ represents spatial motion anchors that localize candidate articulation points. It is updated via a residual scheme $\mathbf{Q}_p^l = \mathbf{Q}_p^{l-1} + \Delta_p^l$, allowing progressive refinement of spatial motion anchors. In parallel, the content query is also updated in a residual manner $\mathbf{Q}_c^l = \mathbf{Q}_c^{l-1} + \Delta_c^l$, where Δ_c^l captures semantic refinement derived from self- and cross-attention. Based on the refined content queries, we predict per-query part class logits $\mathbf{S}_p^l$, which provide structural supervision and are further used to retrieve semantic prototypes from frozen CLIP text embeddings. The retrieved prototypes are fused into the content branch to enhance part-level semantic consistency.

Query Confidence Estimation. Since the number of dual queries N_q defines an upper bound on articulated components, some queries may correspond to invalid part hypotheses. To address this, we predict a confidence score $c_i \in [0, 1]$ from the refined content embedding $\mathbf{Q}_{c,i}^L$ of the i-th query, indicating the reliability of its part hypothesis. During training, confidence supervision is derived